

Spherical Astronomy, Coordinates, and Time

Problem Set

fizicyst

This problem set accompanies the *Spherical Astronomy, Coordinates, and Time* handout. Section A has five objective (multiple-choice) questions; Section B has three subjective questions, pitched a little harder than Section A. Take Earth's radius $R_{\oplus} = 6371$ km and the obliquity of the ecliptic $\epsilon = 23.5^\circ$ where needed, unless a question tells you otherwise.

Section A — Objective

O1. A star has right ascension $\alpha = 5^{\text{h}}30^{\text{m}}$ and, at some moment, hour angle $H = 2^{\text{h}}15^{\text{m}}$. The observer's local sidereal time at that moment is:

- (a) $3^{\text{h}}15^{\text{m}}$
- (b) $7^{\text{h}}45^{\text{m}}$
- (c) $2^{\text{h}}15^{\text{m}}$
- (d) $5^{\text{h}}30^{\text{m}}$

O2. An observer standing exactly at the Earth's north pole sees the celestial equator:

- (a) passing through the zenith
- (b) running along the horizon
- (c) at the nadir
- (d) nowhere — it never appears in the sky from the pole

O3. A star has declination $\delta = 70^\circ\text{N}$, observed from latitude $\phi = 40^\circ\text{N}$. Its altitude at upper culmination is closest to:

- (a) 60°
- (b) 30°
- (c) 50°
- (d) 20°

O4. Which one of the following statements about spherical triangles is **false**?

- (a) The three interior angles of a spherical triangle always sum to more than 180° .
- (b) The area of a spherical triangle is proportional to its angular excess, $E = A + B + C - 180^\circ$.
- (c) The sides of a spherical triangle are arcs of great circles.
- (d) A spherical triangle's angles can sum to exactly 180° , provided it's drawn on a large enough sphere.

O5. An observer stands atop a cliff of height $x = 45$ m above sea level. The dip angle Θ (the extra angle of sight gained from this height) is closest to:

- (a) $13'$
- (b) $130'$
- (c) $1.3'$
- (d) $1300'$

Section B — Subjective

S1. A star has declination $\delta = +35^\circ$ and right ascension $\alpha = 6^{\text{h}}40^{\text{m}}$. It is observed from latitude $\phi = 20^\circ\text{N}$.

- (i) Find the star's altitude at transit ($H = 0$).
- (ii) Find the hour angle H at which the star's altitude is exactly 0° (i.e. at rising/setting). Using the criteria of Section 6.3, confirm this star is neither circumpolar nor a non-riser, i.e. that it rises and sets normally.
- (iii) Taking H negative for rising (east of the meridian, before transit), find the observer's LST at the instant the star rises.

S2. The general altitude formula (Section 12.1.1) is $\sin h = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H$.

- (i) A circumpolar star crosses the local meridian twice a day: once above the pole (upper culmination, $H = 0^\circ$) and once below it (lower culmination, $H = 180^\circ$). Substitute each of these into the general altitude formula to derive expressions for h_u and h_l purely in terms of ϕ and δ . (*Hint:* use $\cos(x \mp y) = \cos x \cos y \pm \sin x \sin y$.)
- (ii) Confirm that your two expressions match $h_u = 90^\circ - |\phi - \delta|$ and $h_l = |\phi + \delta| - 90^\circ$ from Section 6.2 (you may assume $\phi > 0$ and that the star is genuinely circumpolar, so both quantities come out positive).
- (iii) Using your two formulas from (i)–(ii) as a 2×2 linear system in ϕ and δ , show algebraically that ϕ and δ can always be recovered uniquely from a measured h_u and h_l , without simply repeating Section 6.2's specific add/subtract steps — i.e. argue why that system is guaranteed to be invertible.

S3. On a certain day, the Sun's ecliptic longitude is $\lambda_{\odot} = 60^{\circ}$.

- (i) Find the Sun's declination $\delta_{\odot}$ on this day, using $\sin \delta_{\odot} = \sin \epsilon \sin \lambda_{\odot}$.
- (ii) Find the Sun's right ascension $\alpha_{\odot}$ (in hours and minutes), using $\tan \alpha_{\odot} = \cos \epsilon \tan \lambda_{\odot}$ — being careful to place $\alpha_{\odot}$ in the quadrant matching $\lambda_{\odot}$.
- (iii) For an observer at latitude $\phi = 35^{\circ}\text{N}$, use $\delta_{\odot}$ from (i) to find the hour angle of sunset, and hence the total length of daylight (in hours) on this day.